

Tight orthoplectic frames decompose into mutually unbiased tight subframes

Candidate structural partial-solution packet for arXiv:1703.01787

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Status: candidate partial result, likely valid. In the orthoplectic regime, equality of the ordinary and tight Grassmannian constants has an exact structural certificate: a tight optimizer splits canonically into mutually unbiased unit-norm tight subframes. This yields new cardinality bounds and classifies the maximal-cardinality case, but does not decide existence for every pair (n, m) .

1 Question and statement

Source metadata. John I. Haas IV and Peter G. Casazza, *On the structures of Grassmannian frames*, arXiv:1703.01787 (2017), p. 3, Section II.B, Question 2 [1].

and

$$\mu(\Phi) = \mu_{6,3}(\mathbb{R}) = \overline{\mu}_{6,3}(\mathbb{R}) = W_{6,3} = \frac{1}{\sqrt{5}}.$$

Using a combination of analytic and exhaustive methods, the authors of [18] deduced the following two key observations. *Observation (i)* The Grassmannian constant for five unit vectors in $\mathbb{R}^3$ is the same as six vectors in $\mathbb{R}^3$. That is,

$$\overline{\mu}_{5,3}(\mathbb{R}) = \overline{\mu}_{6,3}(\mathbb{R}) = \frac{1}{\sqrt{5}}.$$

Observation (ii) There are no tight frames consisting of five unit vectors in $\mathbb{R}^3$ with coherence that achieves this constant.

Hence, discarding a single vector from Φ yields a new Grassmannian frame; however, Observation (ii) shows that a Grassmannian frame is not tight and, in fact, that any tight frame in $\mathbb{R}^3$ composed of five unit vectors must have strictly greater coherence. In fact, the authors [18] had proven

bound in Theorem 1, Theorem 3 does not imply that an OGF is necessarily tight. In other words, every OGF is a Grassmannian frame but not necessarily a 1-Grassmannian frame. Indeed, for most OGFs belonging to the families of universal Grassmannian frames just mentioned, it is possible to delete several vectors such that the remaining subset is still an OGF [26]; however, the tightness property is almost always lost in these cases. This leads to several questions, but we focus on one.

Question 2. *Given the existence of an OGF $\Phi \in \Omega_{n,m}(\mathbb{F})$, under what conditions does $\mu_{n,m}(\mathbb{F}) = \overline{\mu}_{n,m}(\mathbb{F})$?*

Although we are far from a complete answer to this question, an example from a recent work [29] yields a surprising insight, which we interpret as a partial answer. In [29], the authors construct two distinct orthoplectic Grassmannian frames $\Phi \in \Omega_{n,m}(\mathbb{F})$, where Φ is tight while Ψ is not. Thus

Figure 1: The source’s Question 2 on p. 3. The crop preserves the full page width and the complete question statement.

Source question (transcribed). “Given the existence of an OGF $\Phi \in \Omega_{n,m}(\mathbb{F})$, under what conditions does $\mu_{n,m}(\mathbb{F}) = \overline{\mu}_{n,m}(\mathbb{F})$?” In the source notation the unbarred constant is minimized over unit-norm tight frames and the barred constant over all unit-norm frames.

Let $g_{n,m}(\mathbb{F})$ denote the minimum coherence among all n -element unit-norm frames for $\mathbb{F}^m$, and let $t_{n,m}(\mathbb{F})$ be the corresponding minimum under the additional tightness constraint. These are respectively the Grassmannian and 1-Grassmannian constants of Haas and Casazza [1]. Put

$$d_{\mathbb{F}}(m) = \begin{cases} m(m+1)/2, & \mathbb{F} = \mathbb{R}, \\ m^2, & \mathbb{F} = \mathbb{C}. \end{cases}$$

In the orthoplectic regime $n > d_{\mathbb{F}}(m)$, existence of an orthoplectic Grassmannian frame (OGF) means $g_{n,m}(\mathbb{F}) = 1/\sqrt{m}$. Question 2 of [1] asks when $t_{n,m}(\mathbb{F}) = g_{n,m}(\mathbb{F})$.

Two frames are called *mutually unbiased* here if every cross inner product has modulus $1/\sqrt{m}$.

Proof Intuition

The normalized traceless projector of each frame vector converts the orthoplex coherence inequality into a nonpositive-inner-product condition. Tightness becomes the assertion that these projector vectors sum to zero. Join two projector vectors when their inner product is strictly negative. Different connected components are orthogonal, so the global zero sum splits into a separate zero sum on every component. Translating back says that each component is itself a unit-norm tight frame, while orthogonality between components says precisely that the blocks are mutually unbiased. Inside a connected component the projector Gram matrix is a weighted graph Laplacian; its rank is therefore the block size minus one. The finite dimension of the traceless-matrix space then gives the component and cardinality bounds. The argument also reverses directly. What remains open is the design-theoretic existence of the required blocks for individual parameters, including the complete-MUB obstruction at the complex upper endpoint.

Theorem 1 (Canonical decomposition criterion). *Assume $n > d_{\mathbb{F}}(m)$ and that an OGF exists. Then*

$$t_{n,m}(\mathbb{F}) = g_{n,m}(\mathbb{F}) = \frac{1}{\sqrt{m}} \quad (1)$$

if and only if there is a partition $\Phi = \bigsqcup_{a=1}^r \Phi_a$ of n unit vectors such that

1. *each Φ_a is a unit-norm tight frame for $\mathbb{F}^m$;*
2. *distinct Φ_a, Φ_b are mutually unbiased; and*
3. *every inner product within each Φ_a has modulus at most $1/\sqrt{m}$.*

The partition may be chosen canonically so that the graph on each Φ_a whose edges are the pairs with modulus strictly less than $1/\sqrt{m}$ is connected. For this canonical partition,

$$|\Phi_a| \geq m, \quad r \geq n - d_{\mathbb{F}}(m) + 1, \quad n - r \leq d_{\mathbb{F}}(m) - 1. \quad (2)$$

Thus equality of the two constants is not merely the presence of some tight minimizer: beyond the Gerzon threshold, that minimizer must break into at least two mutually unbiased tight designs.

2 Proof

For a unit vector $\phi \in \mathbb{F}^m$, put $P_\phi = \phi\phi^*$ and use the normalized traceless embedding

$$Q_\phi = \sqrt{\frac{m}{m-1}} \left(P_\phi - \frac{1}{m} I_m \right). \quad (3)$$

The vectors Q_ϕ lie on the unit sphere of the real Hilbert space of traceless symmetric or Hermitian matrices, whose dimension is $D = d_{\mathbb{F}}(m) - 1$. Direct calculation gives

$$\langle Q_\phi, Q_\psi \rangle = \frac{m|\langle \phi, \psi \rangle|^2 - 1}{m-1}. \quad (4)$$

Moreover a unit-norm family Φ is tight precisely when

$$\sum_{\phi \in \Phi} Q_\phi = 0. \quad (5)$$

This zero-sum observation is recorded in [2]; we exploit the sign in (4) as well.

Suppose first that (1) holds. Compactness of the unit-norm-tight-frame manifold gives a tight minimizer Φ with coherence $1/\sqrt{m}$. Write $Q_i = Q_{\phi_i}$. Equations (4) and (5) say

$$\langle Q_i, Q_j \rangle \leq 0 \quad (i \neq j), \quad \sum_i Q_i = 0. \quad (6)$$

Make a graph on the indices by joining i and j exactly when $\langle Q_i, Q_j \rangle < 0$, and let $C_1, \dots, C_r$ be its connected components. Vectors belonging to different components are orthogonal. Consequently their spans are mutually orthogonal. If $S_a = \sum_{i \in C_a} Q_i$, then $S_a \perp S_b$ for $a \neq b$, while $\sum_a S_a = 0$ by (6). Hence

$$0 = \left\| \sum_a S_a \right\|^2 = \sum_a \|S_a\|^2,$$

so every $S_a = 0$. Undoing (3) gives

$$\sum_{i \in C_a} P_{\phi_i} = \frac{|C_a|}{m} I_m. \quad (7)$$

Thus each block $\Phi_a = \{\phi_i : i \in C_a\}$ is a unit-norm tight frame, and in particular $|C_a| \geq m$. Cross-component orthogonality in (4) is exactly mutual unbiasedness. This proves the structural assertions.

It remains to obtain the count. Let G_a be the Gram matrix of the Q_i with $i \in C_a$. It has diagonal entries one, nonpositive off-diagonal entries, and $G_a \mathbf{1} = 0$ because $S_a = 0$. With weights $w_{ij} = -\langle Q_i, Q_j \rangle$, the matrix G_a is therefore the weighted graph Laplacian of the connected graph on C_a . Its kernel is $\text{span}\{\mathbf{1}\}$, so

$$\dim \text{span}\{Q_i : i \in C_a\} = \text{rank } G_a = |C_a| - 1.$$

The component spans are mutually orthogonal inside a D -dimensional space; hence

$$n - r = \sum_a (|C_a| - 1) \leq D = d_{\mathbb{F}}(m) - 1,$$

which is (2).

Conversely, suppose a partition satisfying (1)–(3) of the theorem is given. The union is unit norm and tight because the block frame operators in (7) add to $(n/m)I_m$. Its coherence is at most $1/\sqrt{m}$. Since $n > d_{\mathbb{F}}(m)$, the orthoplex bound says that no unit-norm frame can have smaller coherence. The union is therefore both a tight OGF and a tight-constrained minimizer, proving (1). $\square$

3 Consequences

Corollary 2 (Cardinality obstruction). *If the two constants agree in the orthoplex regime, then*

$$n \leq \begin{cases} \frac{m(m+2)}{2}, & \mathbb{F} = \mathbb{R}, \\ m(m+1), & \mathbb{F} = \mathbb{C}. \end{cases} \quad (8)$$

At the complex upper endpoint $n = m(m+1)$, every tight OGF is a union of $m+1$ mutually unbiased orthonormal bases. At the real upper endpoint, when m is even, every tight OGF is a union of $m/2 + 1$ mutually unbiased orthonormal bases.

Proof. Since every block has at least m elements, $r \leq n/m$. Combine this with $r \geq n - d_{\mathbb{F}}(m) + 1$ to obtain

$$n \leq \frac{m(d_{\mathbb{F}}(m) - 1)}{m - 1},$$

which simplifies to (8). At either stated integral endpoint, equality forces equality in all counting inequalities, so every block has exactly m unit vectors. A unit-norm tight frame of m vectors in $\mathbb{F}^m$ is an orthonormal basis. $\square$

For example, in $\mathbb{C}^2$ a tight five-vector OGF must have exactly two canonical blocks of sizes 2 and 3: an orthonormal basis and a three-vector unit-norm tight frame, mutually unbiased to one another. A tight six-vector OGF must be the union of three mutually unbiased bases. This recovers and structurally explains the tight five-vector example in the papers rather than merely verifying its matrix entries.

4 Verification notes and review recommendation

The projector identity (4) follows by expanding $\text{tr}((P_{\phi} - I/m)(P_{\psi} - I/m))$, and (5) is exactly the tight-frame-operator identity after the same affine change of variables. Compactness of the unit-norm tight-frame manifold ensures that the constrained minimum used in the forward direction is attained. The key splitting step uses only Pythagoras: mutually orthogonal component sums whose total is zero must each vanish. For a connected component, G_a has diagonal one, entries $-w_{ij}$ off the diagonal, and zero row sums, hence is the weighted Laplacian of the strict-edge graph and has one-dimensional kernel. Summing the mutually orthogonal component ranks gives (2); the converse uses only the block frame-operator sum and the orthoplex lower bound. No computational or numerical assertion is used.

Human-review recommendation. Review as a likely-valid substantial partial answer. The highest-value checks are the Laplacian identification and the claim that no equivalent decomposition/cardinality theorem already appears in the OGF or mutually-unbiased-frame literature.

5 Scope, upgrade barrier, and novelty check

The theorem is an exact structural criterion, but not a parameter-only classification: existence of the required mutually unbiased tight subframes remains a separate design problem. At maximal complex cardinality it is precisely the complete-MUB existence problem; for example, a universal parameter theorem in dimension six would decide the famous question of seven complex MUBs. This is why the packet is classified as a strong partial answer rather than a complete numerical table for all (n, m) .

Cheap run indexes contained no prior result for arXiv:1703.01787. Bounded searches through 11 August 2026 for tight OGF decompositions, the cardinality bounds in (8), negative-inner-product graphs, and mutually unbiased tight frames found the source papers and later work introducing mutually unbiased frames [4], but no statement of the converse decomposition theorem above. This is not an exhaustive novelty claim.

References

- [1] J. I. Haas IV and P. G. Casazza, *On the structures of Grassmannian frames*, [arXiv:1703.01787](#) (2017).
- [2] P. G. Casazza and J. I. Haas, *On the rigidity of geometric and spectral properties of Grassmannian frames*, [arXiv:1605.02012](#) (2016).
- [3] B. G. Bodmann and J. I. Haas, *Achieving the orthoplex bound and constructing weighted complex projective 2-designs with Singer sets*, *Linear Algebra Appl.* 511 (2016), 54–71, [doi:10.1016/j.laa.2016.09.005](#).
- [4] F. Caro Pérez, V. González Avella, and D. Goyeneche, *Mutually unbiased frames*, [arXiv:2110.08293](#) (2022).